

GANDAKI UNIVERSITY

Faculty of Science and Technology
Bachelor of Computer Application (BCA)

WEB TECHNOLOGY

CHAPTER 2

HTML AND XHTML

Complete Theory, Examples & Exam Q&A

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2.1 Structuring Documents for the Web

2.1.1 What is HTML?

Definition of HTML

HTML stands for HyperText Markup Language. It is the standard language used to create and design pages on the World Wide Web (WWW). HTML is NOT a programming language — it is a markup language, which means it uses a set of markup tags to describe the structure and content of a web page.

The term **HyperText** refers to text that contains links to other texts (hyperlinks). The term **Markup** refers to annotations added to a document that provide instructions about how to display or structure the document.

Key Historical Points:

- HTML was invented by Tim Berners-Lee in 1991 while working at CERN, Switzerland.
- The first publicly available description was 'HTML Tags', proposed in late 1991.
- HTML 2.0 was released in 1995 as the first standard.
- HTML 4.01 was a widely used version released in 1999.
- XHTML 1.0 was released in 2000 as an XML-based rewrite of HTML 4.01.
- HTML5 was published as a W3C Recommendation in October 2014 and is the current standard.

Characteristics of HTML:

- HTML files are plain text files with .html or .htm extension.
- HTML uses predefined tags enclosed in angle brackets: < and >.
- HTML is platform-independent — it runs on any OS and browser.
- HTML is interpreted by web browsers (Chrome, Firefox, Safari, etc.).
- HTML is case-insensitive in HTML4/HTML5 (but XHTML requires lowercase).
- HTML documents are rendered visually by the browser.

2.1.2 What is XHTML?

Definition of XHTML

XHTML stands for eXtensible HyperText Markup Language. It is a reformulation of HTML 4.01 using the rules of XML (eXtensible Markup Language). XHTML is stricter, cleaner, and more portable than HTML. It was introduced by W3C in January 2000.

XHTML was created to address the problem that HTML browsers were too forgiving — they would render incorrectly written HTML without showing errors. XHTML enforces strict rules so that documents are well-formed and valid XML, making them readable by any XML-aware device or application.

Why XHTML was introduced:

- HTML was becoming too lenient and inconsistent across browsers.
- The rise of mobile and non-traditional browsers required stricter document standards.
- XML-based documents are easier to process by automated tools.
- XHTML ensures backward compatibility while being forward-looking.

2.1.3 Detailed Comparison: HTML vs XHTML

HTML	XHTML
Tags can be uppercase or lowercase: <P>, <p>, <P>	All tags must be lowercase: only <p> is valid
Tags can remain unclosed: , <p>Text	All tags must be closed: , Text
Attribute values can be unquoted: width=200	All attribute values must be quoted: width="200"
Empty elements have no closing: 	Empty elements must self-close:
DOCTYPE is optional	DOCTYPE is mandatory and must appear at top
Attribute minimization allowed: <input checked>	Must use full form: <input checked='checked' />
Nesting errors are tolerated by browsers	Improper nesting causes parsing errors
<html>, <head>, <body> can be omitted	All required elements must be present
Not valid XML	Is valid XML — can be processed by XML parsers
Less strict — older browsers handle errors	Strict — errors must be fixed

XHTML Rules Summary (Must Remember for Exam)

1. All tags must be in lowercase.
2. All tags must be closed (self-closing for void elements).
3. All attribute values must be in double quotes.
4. Attributes must not be minimized.
5. DOCTYPE declaration is mandatory.
6. The root element must be <html>.
7. Elements must be properly nested.

2.1.4 Basic HTML Document Structure (with full explanation)

Every HTML document follows a standard structure. Understanding this structure is fundamental to web development.

```
<!DOCTYPE html>
<html lang='en'>
  <head>
    <meta charset='UTF-8'>
    <meta name='viewport' content='width=device-width, initial-scale=1.0'>
    <meta name='description' content='Web Technology Notes - Gandaki
University'>
    <meta name='author' content='Mr. Dammar Khadayat'>
    <title>Web Technology - Chapter 2</title>
    <link rel='stylesheet' href='style.css'>
  </head>
  <body>
    <h1>Welcome to Web Technology</h1>
    <p>This is the first paragraph of the page.</p>
    <p>HTML structures all web content.</p>
  </body>
</html>
```

Explanation of Each Part:

Element	Purpose / Description
<!DOCTYPE html>	Document Type Declaration. Tells the browser this is an HTML5 document. Must be the very first line. Not an HTML tag itself.
<html lang='en'>	The root element. Every HTML document has exactly one <html> tag. The lang attribute specifies the language (en = English).
<head>	Contains metadata — information about the page that is NOT displayed. Includes title, character set, CSS links, scripts, and meta tags.
<meta charset='UTF-8'>	Specifies the character encoding. UTF-8 supports all languages including Nepali, Hindi, Chinese, Arabic etc.
<meta name='viewport'>	Crucial for responsive design — tells mobile browsers how to scale the page.
<title>	Sets the text shown on the browser tab/title bar. Also used by search engines.

Element	Purpose / Description
<link rel='stylesheet'>	Links an external CSS file to style the HTML page.
<body>	Contains all visible content of the web page: text, images, links, tables, forms, etc.
<h1>	Main heading — displayed as large bold text. There should be only one <h1> per page.
<p>	Paragraph element — creates a block of text with spacing above and below.

2.1.5 HTML Tags, Elements, and Attributes

Tags:

A tag is a keyword (element name) enclosed inside angle brackets. Tags usually come in pairs:

- Opening tag: <tagname> — marks the start of an element
- Closing tag: </tagname> — marks the end (note the forward slash /)
- Self-closing tag: <tagname /> — for empty elements that have no content

Elements:

An HTML element is everything from the opening tag to the closing tag, including the content in between.

```
<p>This is a paragraph element.</p>
  ^opening tag ^content      ^closing tag

<br />    <!-- Self-closing / void element - has no content -->
<img src='photo.jpg' alt='Photo' /> <!-- void element with attributes -->
```

Attributes:

Attributes provide additional information about an element. They are always placed inside the opening tag and have a name='value' format.

```
<a href='https://www.gandaki.edu.np' target='_blank' title='Gandaki
University'>
  Visit Gandaki University
</a>

<!-- Attributes: href   = URL destination -->
<!--               target = how to open -->
<!--               title  = tooltip text -->
```

Attribute	Used With	Purpose
href	<a>, <link>	Specifies the URL/destination of a link
src	, <script>, <audio>, <video>	Specifies the path/URL of the resource
alt		Alternative text for accessibility and SEO
id	Any element	Unique identifier — used by CSS and JavaScript
class	Any element	Class name for CSS styling — multiple elements can share a class
style	Any element	Inline CSS styles applied directly to the element
name	<input>, <form>, <meta>	Name identifier for form data submission
type	<input>, <button>, <script>	Specifies the type of element
width / height	, <table>	Sets dimensions in pixels
colspan	<td>, <th>	Number of columns a cell spans
rowspan	<td>, <th>	Number of rows a cell spans
action	<form>	URL where form data is submitted
method	<form>	HTTP method: GET or POST
placeholder	<input>	Hint text inside an empty input field
required	<input>	Makes a field mandatory before form submission

Global vs Local Attributes:

- Global Attributes: Can be used with ANY HTML element — id, class, style, title, lang, dir, tabindex, hidden.
- Local Attributes: Specific to certain elements — href (for <a>), src (for), action (for <form>).

2.1.6 Comments in HTML

Comments are notes added to HTML code for documentation. They are not displayed in the browser.

```
<!-- This is a single-line comment -->

<!--
  This is a multi-line comment.
  It can span multiple lines.
  Useful for documenting complex sections.
-->
```

```
-->

<!-- TODO: Add navigation bar here -->
<nav> ... </nav>
```

Common Mistake

HTML comments CANNOT be nested. <!-- outer <!-- inner --> outer end --> will cause errors. Comments also cannot contain double hyphens (--) inside them.

2.2 Text Formatting — Presentational and Phrase Elements

2.2.1 Headings

HTML provides six levels of headings from `<h1>` (largest, most important) to `<h6>` (smallest, least important). Headings are block-level elements — they start on a new line and add space above and below.

```
<h1>Heading Level 1 – Main Title (Largest)</h1>
<h2>Heading Level 2 – Section Title</h2>
<h3>Heading Level 3 – Sub-section</h3>
<h4>Heading Level 4</h4>
<h5>Heading Level 5</h5>
<h6>Heading Level 6 – Smallest</h6>
```

Best Practice for Headings

Use only ONE `<h1>` per page (for SEO and accessibility). Use headings in order — don't skip from `<h1>` to `<h4>`. Search engines use headings to understand page structure. Screen readers use headings for navigation.

2.2.2 Paragraphs and Line Breaks

```
<p>This is the first paragraph. The browser automatically
adds space above and below a paragraph element.</p>
```

```
<p>This is the second paragraph. Multiple spaces   and
line breaks in the source code are collapsed to one space
by the browser.</p>
```

```
<p>To force a line break<br />inside a paragraph,<br />
use the br tag.</p>
```

```
<hr />  <!-- Horizontal Rule: draws a dividing line -->
```

```
<pre>
  This text preserves   all spaces
  and line breaks exactly as typed.
  Used for code display or ASCII art.
</pre>
```

2.2.3 Text Formatting Tags — Complete Reference

Tag	Category	Visual Effect	Semantic Meaning	Example
	Presentation al	Bold	None — purely visual	Bold text
<i>	Presentation al	Italic	None — purely visual	<i>Italic text</i>
<u>	Presentation al	Underline	None — purely visual	<u>Underlined</u>
<s> or <strike>	Presentation al	Strikethrough	None — deprecated	<s>Old text</s>
	Phrase	Bold	Strong importance	Warning!
	Phrase	Italic	Emphasis / stress	Key word
<mark>	Phrase	Yellow highlight	Highlighted/relevant	<mark>Found text</mark>
	Phrase	Strikethrough	Deleted content	Removed
<ins>	Phrase	Underline	Inserted content	<ins>Added text</ins>
<small>	Phrase	Smaller font	Fine print	Copyright 2024
<big>	Presentation al	Larger font	None — deprecated	<big>Big text</big>
<sub>	Phrase	Subscript	Subscript notation	H₂O
<sup>	Phrase	Superscript	Superscript notation	x²+y²
<code>	Phrase	Monospace font	Computer code	<code>print(x)</code>
<kbd>	Phrase	Monospace font	Keyboard input	Press <kbd>Ctrl+C</kbd>
<samp>	Phrase	Monospace font	Sample output	<samp>Error 404</samp>
<var>	Phrase	Italic	Variable name	<var>x</var> = 5
<abbr>	Phrase	Dotted underline	Abbreviation	<abbr title='HyperText'>HTML</abbr>
<cite>	Phrase	Italic	Citation/reference	<cite>Web Technology Book</cite>
<blockquote>	Block	Indented block	Long quotation	<blockquote>Long quote...</blockquote>

Tag	Category	Visual Effect	Semantic Meaning	Example
<code><q></code>	Inline	Adds quotation marks	Inline quotation	<code><q>Short quote</q></code>
<code><address></code>	Block	Italic	Contact information	<code><address>Pokhara, Nepal</address></code>

2.2.4 Presentational vs Phrase Elements — Deep Comparison

This distinction is frequently asked in exams. Understanding the conceptual difference is essential.

Presentational Elements	Phrase Elements (Semantic)
Deal ONLY with visual appearance	Carry meaning/semantic information about content
Provide no information to browsers, search engines, or screen readers	Convey structural meaning to browsers, search engines, and assistive technology
Examples: <code></code> , <code><i></code> , <code><u></code> , <code><big></code> , <code><center></code> , <code></code>	Examples: <code></code> , <code></code> , <code><mark></code> , <code></code> , <code><ins></code> , <code><abbr></code>
Deprecated or discouraged in HTML5	Preferred and recommended in HTML5
Screen readers ignore them — no accessibility benefit	Screen readers interpret them — improves accessibility
<code>Important</code> — just looks bold	<code>Important</code> — also means 'this is important'
CSS alternative: <code>font-weight: bold;</code>	CSS can change look but semantic meaning remains

Practical Example of the Difference

Consider: `<b>Warning</b>` vs `<strong>Warning</strong>`

Both display as bold text in a browser.

However, `<strong>` tells a screen reader to read the word with more vocal emphasis, which helps visually impaired users.

`<b>` just makes the text appear bold — the screen reader reads it normally.

Conclusion: Always use `<strong>` instead of `<b>`, and `<em>` instead of `<i>`, for semantic HTML.

2.3 Lists, Links, Images, Audio, and Video

2.3.1 Lists in HTML

Lists are used to group related items together. HTML provides three types of lists. Lists can also be nested inside each other.

A. Unordered List

Used for items where order does NOT matter. Items appear with bullet points by default. The type attribute (or CSS) can change the bullet style.

```
<ul>
  <li>Computer Science</li>
  <li>Web Technology</li>
  <li>Database Management</li>
  <li>Networking</li>
</ul>

<!-- Using type attribute (HTML4, now use CSS) -->
<ul type='disc'>  <!-- filled circle (default) -->
<ul type='circle'> <!-- hollow circle -->
<ul type='square'> <!-- filled square -->
```

B. Ordered List

Used when the order of items MATTERS (steps, rankings, sequences). Items are automatically numbered.

```
<ol>
  <li>Open the text editor (Notepad++, VS Code)</li>
  <li>Write the HTML code</li>
  <li>Save the file as index.html</li>
  <li>Open the file in a web browser</li>
  <li>View the result</li>
</ol>

<!-- type attribute controls numbering style -->
<ol type='1'>  <!-- 1, 2, 3 (default) -->
<ol type='A'>  <!-- A, B, C -->
<ol type='a'>  <!-- a, b, c -->
<ol type='I'>  <!-- I, II, III (uppercase Roman) -->
<ol type='i'>  <!-- i, ii, iii (lowercase Roman) -->

<!-- start attribute sets the starting number -->
<ol start='5'> <!-- starts from 5, 6, 7 ... -->
```

C. Definition List <dl>

Used for glossaries, dictionaries, or any term-definition pairs. Uses three tags: <dl> (definition list), <dt> (definition term), <dd> (definition description).

```
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language. The standard language for
    creating web pages and web applications.</dd>

  <dt>CSS</dt>
  <dd>Cascading Style Sheets. A language for describing the
    presentation of HTML documents.</dd>

  <dt>JavaScript</dt>
  <dd>A programming language that enables dynamic web content,
    client-side interaction, and web applications.</dd>
</dl>
```

D. Nested Lists

Lists can be placed inside other lists to create hierarchical/multilevel structures.

```
<ul>
  <li>Computer Science Subjects
    <ul>
      <li>Programming
        <ol>
          <li>C Programming</li>
          <li>Java</li>
          <li>Python</li>
        </ol>
      </li>
      <li>Database Systems</li>
    </ul>
  </li>
  <li>Mathematics</li>
</ul>
```

Unordered List	Ordered List
Items have NO specific order	Items follow a specific sequence
Displayed with bullet points	Displayed with numbers or letters
Used for: navigation menus, feature lists	Used for: instructions, rankings, procedures
type: disc, circle, square	type: 1, A, a, I, i
No start attribute	start attribute to begin at custom number

2.3.2 Hyperlinks — The <a> Anchor Tag

What is a Hyperlink?

A hyperlink is a clickable reference that allows users to navigate from one web page to another, to a different section of the same page, to an email address, or to download a file. Hyperlinks are created using the <a> (anchor) tag.

Syntax:

```
<a href='destination'>Link Text</a>
```

Types of Links with Examples:

```
<!-- 1. External Link - links to another website -->
<a href='https://www.gandaki.edu.np'>Gandaki University</a>

<!-- 2. Internal Link - links to another page on same site -->
<a href='about.html'>About Us</a>
<a href='../contact/contact.html'>Contact</a>    <!-- parent folder -->

<!-- 3. Anchor Link - jumps to a section on same page -->
<a href='#section2'>Go to Section 2</a>    <!-- link -->
<h2 id='section2'>Section 2 Content</h2>    <!-- target -->

<!-- 4. Email Link - opens default email client -->
<a href='mailto:khadayat@gandaki.edu.np'>Email Us</a>

<!-- 5. Phone Link - useful on mobile devices -->
<a href='tel:+977-61-123456'>Call Us</a>

<!-- 6. Download Link - forces file download -->
<a href='notes.pdf' download='Chapter2_Notes.pdf'>Download Notes</a>

<!-- 7. Image as Link - wraps an image in an anchor tag -->
<a href='home.html'>
  <img src='logo.png' alt='Home' width='100' />
</a>
```

The target Attribute — Where to Open the Link:

Value	Behavior	Use Case
_self	Opens in the same tab (DEFAULT behavior)	Normal page navigation
_blank	Opens in a new browser tab or window	External websites, PDFs, downloads

Value	Behavior	Use Case
_parent	Opens in the parent frame (in framesets)	Used with HTML frames
_top	Opens in the full browser window, breaking out of frames	Breaking out of nested frames
framename	Opens in a named <iframe> or frame	Targeting a specific frame on the page

Security Note

When using target='_blank', always add rel='noopener noreferrer' to prevent security vulnerabilities. Example: Link

2.3.3 Images — The Tag

The tag is used to embed images into web pages. It is an empty/void element — it has no closing tag. In XHTML, it must be written as .

Complete Syntax with All Attributes:

```
<img
  src='images/campus.jpg'
  alt='Gandaki University Campus Building'
  width='500'
  height='300'
  title='Our Beautiful Campus'
  loading='lazy'
  class='campus-photo'
/>
```

Attribute	Required?	Purpose	Example
src	YES	Source path or URL of the image file	src='photo.jpg' or src='https://...'
alt	YES (XHTML) / Strongly recommended	Alternative text shown if image fails to load. Used by screen readers and SEO.	alt='Student studying in library'
width	Optional	Width of image in pixels. If only width is set, height scales proportionally.	width='400'
height	Optional	Height of image in pixels.	height='300'
title	Optional	Tooltip text shown when user hovers over the image.	title='Campus photo 2024'

Attribute	Required?	Purpose	Example
loading	Optional	lazy = load image only when visible (improves performance). eager = load immediately.	loading='lazy'
class/id	Optional	For CSS styling and JavaScript targeting.	class='profile-pic'

Supported Image Formats:

- JPEG / JPG — Best for photographs. Small file size, lossy compression. Extension: .jpg, .jpeg
- PNG — Best for graphics, logos, images with transparency. Lossless. Extension: .png
- GIF — Supports animation. Limited to 256 colors. Extension: .gif
- SVG — Scalable Vector Graphics. Perfect for logos/icons — scales without losing quality. Extension: .svg
- WebP — Modern format by Google. Smaller size than JPEG/PNG with better quality. Extension: .webp

Using Images as Links (Exam Favourite)

Images can be made clickable by wrapping them inside an <a> tag:

```
<a href="gallery.html"></a>
```

This is commonly used for logos (clicking the logo returns to home page) and thumbnail galleries.

2.3.4 Adding Audio — HTML5 <audio> Tag

Before HTML5, audio could only be embedded using Flash Player or third-party plugins. HTML5 introduced the native <audio> element, which works in all modern browsers without any plugin.

```
<!-- Basic Audio with Controls -->
<audio controls>
  <source src='music.mp3' type='audio/mpeg'>
  <source src='music.ogg' type='audio/ogg'>
  <source src='music.wav' type='audio/wav'>
  Your browser does not support the audio element.
  Please upgrade to a modern browser.
</audio>

<!-- Audio with all attributes -->
<audio
  controls
  autoplay
  loop
```

```

    muted
    preload='auto'
  >
  <source src='national_anthem.mp3' type='audio/mpeg'>
</audio>

```

Attribute	Purpose	Values
controls	Shows the audio player controls (play, pause, volume, progress bar)	No value needed (boolean attribute)
autoplay	Audio starts playing automatically when page loads (often blocked by browsers)	No value needed
loop	Audio repeats continuously in a loop	No value needed
muted	Audio starts muted (silent)	No value needed
preload	Hints to browser how to load the audio before user plays it	auto, metadata, none
src	Can set source directly on <audio> (instead of <source> tags)	URL or file path

Supported Audio Formats:

- MP3 (audio/mpeg) — Most widely supported. Good compression.
- OGG (audio/ogg) — Open source format. Good quality.
- WAV (audio/wav) — Uncompressed, high quality. Large file size.
- AAC (audio/aac) — Better quality than MP3 at same bitrate. Used by Apple.

2.3.5 Adding Video — HTML5 <video> Tag

HTML5 <video> tag embeds video files directly in web pages without Flash or any plugin.

```

<!-- Basic Video Player -->
<video width='640' height='360' controls>
  <source src='lecture.mp4' type='video/mp4'>
  <source src='lecture.webm' type='video/webm'>
  <source src='lecture.ogv' type='video/ogg'>
  Your browser does not support HTML5 video.
</video>

<!-- Video with all attributes -->
<video
  width='800'
  height='450'
  controls
  autoplay

```

```

    loop
    muted
    poster='thumbnail.jpg'
    preload='metadata'
  >
  <source src='promo.mp4' type='video/mp4'>
</video>

```

Attribute	Purpose
width / height	Sets the display dimensions of the video player in pixels
controls	Shows video controls: play/pause, volume, fullscreen, progress bar
autoplay	Video plays automatically (browsers often require muted with autoplay)
loop	Video restarts when it finishes
muted	Video starts without sound
poster	Image file to display as a thumbnail before the video plays or while loading
preload	auto: load full video. metadata: load only info. none: don't preload.

Flash (Old Method)	HTML5 Audio/Video (New Method)
Requires Adobe Flash Player plugin	Native browser support — no plugin needed
Not supported on iOS/iPhone	Works on all devices including mobile
Security vulnerabilities — frequently exploited	More secure — no third-party plugin
Adobe ended Flash support in December 2020	Actively maintained W3C standard
Large file size overhead from Flash	Lightweight, efficient loading
Not accessible to screen readers	Better accessibility support
<object> or <embed> tag required	Simple <audio> or <video> tag

2.4 Tables in HTML

What is an HTML Table?

An HTML table is used to display data in a structured grid format of rows and columns. Tables are ideal for displaying tabular data such as schedules, mark sheets, price lists, comparison charts, and reports. They should NOT be used for page layout (that is the job of CSS).

2.4.1 Core Table Elements

Tag	Full Name	Purpose
<table>	Table	Container element that defines the entire table
<tr>	Table Row	Defines a horizontal row in the table
<th>	Table Header	Defines a header cell. Bold and center-aligned by default.
<td>	Table Data	Defines a regular data cell
<caption>	Caption	Provides a title/description for the table (appears above by default)
<thead>	Table Head	Groups the header row(s) — semantic grouping
<tbody>	Table Body	Groups the main data rows — semantic grouping
<tfoot>	Table Footer	Groups footer row(s) such as totals — semantic grouping
<colgroup>	Column Group	Groups columns for collective styling
<col>	Column	Defines properties for a specific column

2.4.2 Basic Table Structure

```
<table border='1' cellpadding='8' cellspacing='0'>
  <caption>BCA Student Mark Sheet – 2024</caption>

  <thead>
    <tr>
      <th>Roll No</th>
      <th>Student Name</th>
      <th>Web Technology</th>
      <th>Database</th>
      <th>Total</th>
    </tr>
  </thead>
```

```

<tbody>
  <tr>
    <td>101</td>
    <td>Ram Bahadur Sharma</td>
    <td>85</td>
    <td>78</td>
    <td>163</td>
  </tr>
  <tr>
    <td>102</td>
    <td>Sita Kumari Thapa</td>
    <td>90</td>
    <td>88</td>
    <td>178</td>
  </tr>
  <tr>
    <td>103</td>
    <td>Hari Prasad Poudel</td>
    <td>75</td>
    <td>82</td>
    <td>157</td>
  </tr>
</tbody>

<tfoot>
  <tr>
    <td colspan='2'>Class Average</td>
    <td>83.3</td>
    <td>82.7</td>
    <td>166</td>
  </tr>
</tfoot>
</table>

```

Table Attributes Explained:

Attribute	Purpose	Example
border	Sets the width of the border in pixels. 0 = no border.	border='1'
cellpadding	Space between cell content and cell border (inner spacing)	cellpadding='8'
cellspacing	Space between individual cells (gap between cells)	cellspacing='0'
width	Overall width of the table (pixels or percentage)	width='80%' or width='600'
bgcolor	Background color of the table or cell	bgcolor='#CCCCFF'

Attribute	Purpose	Example
align	Horizontal alignment: left, center, right (table or cell)	align='center'
valign	Vertical alignment inside cells: top, middle, bottom	valign='top'

2.4.3 colspan — Spanning Columns

colspan Attribute

The colspan attribute allows a table cell (<td> or <th>) to span across (merge) multiple columns horizontally. It eliminates the need for multiple adjacent cells when a single cell should cover more than one column width.

Syntax: <td colspan='n'> — where n is the number of columns the cell should span.

```
<table border='1' cellpadding='6'>

  <!-- Row 1: Title spanning all 4 columns -->
  <tr>
    <th colspan='4'>Annual Examination Result – Gandaki University
    2024</th>
  </tr>

  <!-- Row 2: Column Headers -->
  <tr>
    <th>Roll No</th>
    <th>Name</th>
    <th colspan='2'>Marks (Theory / Practical)</th>
  </tr>

  <!-- Row 3: Data -->
  <tr>
    <td>201</td>
    <td>Ram Sharma</td>
    <td>75</td>
    <td>22</td>
  </tr>

  <!-- Row 4: Total spanning 3 columns -->
  <tr>
    <td colspan='3'>Grand Total</td>
    <td>97</td>
  </tr>
</table>

<!-- IMPORTANT RULE: If a row has 4 columns, and you use
  colspan='4' in one cell, that cell takes all 4 columns.
  You cannot add more <td> cells in that row. -->
```

colspan Rule

The total column count in every row must be consistent.

If your table has 4 columns and you use `<th colspan='4'>`, that row has only 1 cell (spanning all 4).

If you use `<td colspan='2'>`, you can add only 2 more regular `<td>` cells in that row ($2+2=4$).

Breaking this rule causes the table to display incorrectly.

2.4.4 rowspan — Spanning Rows

rowspan Attribute

The rowspan attribute allows a table cell to span vertically across multiple rows. When a cell uses `rowspan='n'`, it occupies n rows, and the affected rows have one fewer `<td>` or `<th>` cell.

Syntax: `<td rowspan='n'>` — where n is the number of rows the cell should span.

```
<table border='1' cellpadding='6'>
  <tr>
    <th>Student</th>
    <th>Subject</th>
    <th>Marks</th>
  </tr>

  <!-- Ram spans 3 rows (has 3 subjects) -->
  <tr>
    <td rowspan='3'>Ram Sharma</td> <!-- occupies rows 2, 3, 4 -->
    <td>Web Technology</td>
    <td>85</td>
  </tr>
  <tr>
    <!-- No <td> for 'Student' column - occupied by rowspan -->
    <td>Database Systems</td>
    <td>78</td>
  </tr>
  <tr>
    <!-- No <td> for 'Student' column - still occupied -->
    <td>Networking</td>
    <td>82</td>
  </tr>

  <!-- New student -->
  <tr>
    <td rowspan='2'>Sita Thapa</td>
    <td>Web Technology</td>
    <td>90</td>
  </tr>
  <tr>
    <td>Database Systems</td>
    <td>88</td>
  </tr>
```

```
</table>
```

2.4.5 colspan vs rowspan — Final Comparison

colspan	rowspan
Merges cells HORIZONTALLY (across columns)	Merges cells VERTICALLY (across rows)
colspan='n' — spans n columns	rowspan='n' — spans n rows
The cell becomes wider — covers more columns	The cell becomes taller — covers more rows
Other cells in same ROW are reduced by (n-1)	Other cells in subsequent ROWS are reduced by 1
Common use: Table title header, subtotal labels	Common use: Student/category name appearing for multiple rows
Example: <code><th colspan='3'>Student Results</th></code>	Example: <code><td rowspan='4'>Computer Science</td></code>

Combined colspan and rowspan

Both can be used together in the same table. A cell can use both: `<td colspan='2' rowspan='2'>` — this cell spans 2 columns AND 2 rows simultaneously, creating a 2×2 merged cell. This is commonly seen in complex schedule tables and timetables.

2.5 Forms and Form Controls

What is an HTML Form?

An HTML form is a section of a web page that collects user input and submits it to a server for processing. Forms are the primary mechanism for user interaction in web applications — used for login, registration, search, checkout, surveys, and more.

Forms contain various input controls (widgets) such as text boxes, password fields, checkboxes, radio buttons, dropdown lists, file upload buttons, and submit buttons. When the user submits the form, the data is sent to a server-side script (PHP, Python, etc.) for processing.

2.5.1 The <form> Element

Syntax:

```
<form
  action='process.php'
  method='POST'
  id='registrationForm'
  name='regForm'
  enctype='multipart/form-data'
  onsubmit='return validateForm()'
  onreset='confirmReset()'
  target='_self'
  accept-charset='UTF-8'
>
  <!-- Form controls go here -->
</form>
```

Detailed Explanation of Form Attributes:

Attribute	Purpose	Values / Example
action	Specifies the URL of the server-side script that processes the form data. If omitted, data is sent to the same page.	action='register.php' action='https://api.example.com/submit'
method	HTTP method used to send data. GET appends data to URL. POST sends data in request body.	method='GET' method='POST'

Attribute	Purpose	Values / Example
id	Unique identifier for the form element. Used by JavaScript (document.getElementById) and CSS.	id='loginForm'
name	Name of the form for scripting. Used by older JavaScript: document.forms['name'].	name='contactForm'
enctype	Encoding type for form data. MUST be set to multipart/form-data when the form includes file uploads.	enctype='application/x-www-form-urlencoded' (default) enctype='multipart/form-data' (for file upload) enctype='text/plain'
onsubmit	JavaScript function called when user clicks Submit. Return false to prevent submission (for validation).	onsubmit='return validate()'
onreset	JavaScript function called when Reset button is clicked.	onreset='confirmReset()'
target	Where to display the server response after submission.	target='_self' (same tab, default) target='_blank' (new tab)
accept-charset	Character sets the server can accept. UTF-8 is recommended for international support.	accept-charset='UTF-8'
autocomplete	Enables/disables browser autofill for the form.	autocomplete='on' or 'off'
novalidate	Disables HTML5 built-in form validation. Useful when custom JS validation is used.	novalidate (boolean attribute)

2.5.2 GET vs POST — Detailed Comparison

GET Method	POST Method
Form data is appended to the URL as a query string	Form data is sent in the HTTP request body (invisible)
URL: page.php?name=Ram&age=20	URL remains clean: page.php
Data is VISIBLE in the browser address bar	Data is NOT visible in address bar
Data is stored in browser history and can be bookmarked	Data is NOT stored in history and cannot be bookmarked
Limited data size (~2,048 characters)	No practical limit on data size
LESS secure — data visible in URL, logs, referrer headers	MORE secure — data not in URL

GET Method	POST Method
Suitable for: Search forms, filters, public data	Suitable for: Login, registration, payments, file upload
Can be cached by browser	Not cached by browser
Can be resent by refreshing the page	Browser warns before resubmitting
Idempotent — same request gives same result	Not idempotent — may have side effects

When to Use Which Method

Use GET: Search bars, filter forms, public queries — because results can be bookmarked and shared.

Use POST: Login forms, registration, file uploads, payment forms, anything involving passwords or sensitive data.

Rule of thumb: If the form retrieves data → GET. If the form creates/modifies data → POST.

2.5.3 The <input> Element — All Types

The <input> element is the most versatile form control. Its behavior is entirely determined by the type attribute. HTML5 added many new input types.

```
<!-- TEXT — single line text input -->
<input type='text' name='username' placeholder='Enter your name'
       maxlength='50' size='30' required />

<!-- PASSWORD — masked text -->
<input type='password' name='pwd' placeholder='Enter password'
       minlength='8' required />

<!-- EMAIL — validates email format -->
<input type='email' name='email' placeholder='you@example.com' />

<!-- NUMBER — numeric input with range -->
<input type='number' name='age' min='1' max='120' step='1' value='18' />

<!-- DATE — calendar date picker -->
<input type='date' name='dob' min='2000-01-01' max='2024-12-31' />

<!-- RANGE — slider control -->
<input type='range' name='rating' min='1' max='10' step='1' value='5' />

<!-- COLOR — color picker -->
<input type='color' name='favcolor' value='#1F3864' />

<!-- TEL — telephone number -->
```

```

<input type='tel' name='phone' pattern='[0-9]{10}' placeholder='10-digit
number' />

<!-- URL - validates URL format -->
<input type='url' name='website' placeholder='https://...' />

<!-- SEARCH - styled as a search field -->
<input type='search' name='q' placeholder='Search...' />

<!-- FILE - file upload button -->
<input type='file' name='photo' accept='image/*' multiple />

<!-- HIDDEN - invisible, stores data -->
<input type='hidden' name='user_id' value='12345' />

<!-- SUBMIT - submits the form -->
<input type='submit' value='Submit Application' />

<!-- RESET - clears the form -->
<input type='reset' value='Clear All Fields' />

<!-- BUTTON - clickable button -->
<input type='button' value='Preview' onclick='showPreview()' />

<!-- IMAGE - clickable image as submit button -->
<input type='image' src='submit-btn.png' alt='Submit' />

<!-- CHECKBOX - multiple selection -->
<input type='checkbox' name='hobby' value='reading' id='hob1' checked />
<label for='hob1'>Reading</label>

<!-- RADIO - single selection from a group -->
<input type='radio' name='gender' value='male' id='gM' />
<label for='gM'>Male</label>
<input type='radio' name='gender' value='female' id='gF' />
<label for='gF'>Female</label>

```

Checkbox	Radio Button
Allows MULTIPLE selections from a group	Allows only ONE selection per group (same name)
Each checkbox is independent	Radio buttons with same name attribute form a group
User can select 0, 1, or many	User must select exactly one
Submitted as multiple values with same name	Only selected value is submitted
type='checkbox'	type='radio'
Example: Choose hobbies (can pick Reading AND Coding)	Example: Choose gender (only Male OR Female)

Checkbox	Radio Button
Visually: square tick box	Visually: round circle button

2.5.4 Other Form Controls

Textarea — Multi-line Text Input:

```
<label for='msg'>Your Message:</label><br />
<textarea
  id='msg'
  name='message'
  rows='6'
  cols='50'
  placeholder='Type your message here (max 500 characters)...'
  maxlength='500'
  required
>
</textarea>

<!-- rows: visible height in character lines -->
<!-- cols: visible width in average characters -->
```

Select Box — Dropdown Menu:

```
<!-- Basic Select -->
<select name='province'>
  <option value=''>-- Select Province --</option>
  <option value='1'>Province 1 (Koshi)</option>
  <option value='2'>Madhesh Province</option>
  <option value='3'>Bagmati Province</option>
  <option value='4' selected>Gandaki Province</option>
  <option value='5'>Lumbini Province</option>
  <option value='6'>Karnali Province</option>
  <option value='7'>Sudurpashchim Province</option>
</select>

<!-- Multiple Selection (hold Ctrl to select multiple) -->
<select name='subjects' multiple size='4'>
  <option value='wt'>Web Technology</option>
  <option value='db'>Database</option>
  <option value='nw'>Networking</option>
  <option value='os'>Operating Systems</option>
</select>

<!-- Grouped Options with <optgroup> -->
<select name='course'>
  <optgroup label='Undergraduate'>
    <option value='bca'>BCA</option>
    <option value='bsc'>BSc CSIT</option>
```

```
</optgroup>
<optgroup label='Postgraduate'>
  <option value='mca'>MCA</option>
  <option value='msc'>MSc</option>
</optgroup>
</select>
```

File Upload:

```
<!-- Single file upload -->
<input type='file' name='resume' accept='.pdf,.doc,.docx' />

<!-- Multiple files -->
<input type='file' name='photos' accept='image/*' multiple />

<!-- IMPORTANT: Form must have enctype='multipart/form-data' -->
<form action='upload.php' method='POST' enctype='multipart/form-data'>
  <input type='file' name='document' />
  <input type='submit' value='Upload' />
</form>
```

File Upload Critical Requirement

When a form includes a file upload field, the form's enctype attribute MUST be set to 'multipart/form-data'. Without this, only the filename is sent, not the actual file content.

Hidden Fields:

```
<!-- Hidden fields are not visible to users but are submitted with the form -
->
<!-- Used to pass session IDs, CSRF tokens, product IDs, etc. -->
<input type='hidden' name='csrf_token' value='abc123xyz' />
<input type='hidden' name='product_id' value='PRD-00524' />
<input type='hidden' name='form_version' value='2.1' />
```

Labels — For Accessibility:

```
<!-- Method 1: for/id association (preferred) -->
<label for='email'>Email Address:</label>
<input type='email' id='email' name='email' />

<!-- Method 2: wrapping label around input -->
<label>
  Full Name:
  <input type='text' name='fullname' />
</label>

<!-- Clicking the label text focuses/activates the input -->
```

```
<!-- Critically important for accessibility and mobile usability -->
```

2.5.5 Complete Student Registration Form — Full Example

```
<!DOCTYPE html>
<html lang='en'>
<head>
  <meta charset='UTF-8'>
  <title>Student Registration – Gandaki University</title>
  <style>
    body { font-family: Arial, sans-serif; margin: 30px; background:
#f0f4f8; }
    .form-container { background: white; padding: 30px; border-radius: 8px;
max-width: 600px; margin: auto; box-shadow: 0 2px 8px
#ccc; }
    h2 { color: #1F3864; text-align: center; }
    label { display: block; margin-top: 14px; font-weight: bold; color:
#333; }
    input[type='text'], input[type='email'], input[type='tel'],
input[type='date'], select, textarea {
      width: 100%; padding: 8px; border: 1px solid #aaa;
      border-radius: 4px; box-sizing: border-box; margin-top: 4px; }
    .radio-group label, .check-group label { display: inline; margin-right:
16px;
                                font-weight: normal; }
    .btn { margin-top: 20px; padding: 10px 24px; font-size: 15px;
      border: none; border-radius: 4px; cursor: pointer; }
    .btn-submit { background: #1F3864; color: white; }
    .btn-reset { background: #aaa; color: white; margin-left: 10px; }
  </style>
</head>
<body>
<div class='form-container'>
  <h2>Student Registration Form</h2>
  <form action='register.php' method='POST' enctype='multipart/form-data'
    id='regForm' onsubmit='return validateForm()'>

    <!-- Personal Information -->
    <label for='fname'>Full Name: *</label>
    <input type='text' id='fname' name='fullname'
      placeholder='e.g. Ram Bahadur Sharma' required maxlength='100'
  />

    <label for='email'>Email Address: *</label>
    <input type='email' id='email' name='email'
      placeholder='you@example.com' required />

    <label for='phone'>Phone Number:</label>
    <input type='tel' id='phone' name='phone'
      placeholder='98XXXXXXXX' pattern='[0-9]{10}' />

    <label for='dob'>Date of Birth:</label>
```

```
<input type='date' id='dob' name='dob'
      min='2000-01-01' max='2010-12-31' />

<!-- Gender (Radio Buttons) -->
<label>Gender: *</label>
<div class='radio-group'>
  <input type='radio' name='gender' id='gM' value='Male' required />
  <label for='gM'>Male</label>
  <input type='radio' name='gender' id='gF' value='Female' />
  <label for='gF'>Female</label>
  <input type='radio' name='gender' id='gO' value='Other' />
  <label for='gO'>Other</label>
</div>

<!-- Province (Select Box) -->
<label for='province'>Province:</label>
<select id='province' name='province'>
  <option value=''>-- Select Province --</option>
  <option value='Koshi'>Koshi Province</option>
  <option value='Bagmati'>Bagmati Province</option>
  <option value='Gandaki' selected>Gandaki Province</option>
  <option value='Lumbini'>Lumbini Province</option>
</select>

<!-- Hobbies (Checkboxes) -->
<label>Hobbies:</label>
<div class='check-group'>
  <input type='checkbox' name='hobby' id='hReading' value='Reading' />
  <label for='hReading'>Reading</label>
  <input type='checkbox' name='hobby' id='hCoding' value='Coding' />
  <label for='hCoding'>Coding</label>
  <input type='checkbox' name='hobby' id='hSports' value='Sports' />
  <label for='hSports'>Sports</label>
  <input type='checkbox' name='hobby' id='hMusic' value='Music' />
  <label for='hMusic'>Music</label>
</div>

<!-- Photo Upload -->
<label for='photo'>Upload Passport Photo:</label>
<input type='file' id='photo' name='photo'
      accept='image/jpeg,image/png' />

<!-- Address (Textarea) -->
<label for='address'>Permanent Address:</label>
<textarea id='address' name='address' rows='3'
      placeholder='Village/Ward, Municipality, District,
Province'></textarea>

<!-- Hidden fields -->
<input type='hidden' name='form_id' value='REG-2024' />
<input type='hidden' name='semester' value='1' />

<!-- Submit and Reset -->
<div>
```

```
        <button type='submit' class='btn btn-submit'>Submit  
Registration</button>  
        <button type='reset' class='btn btn-reset' >Clear Form</button>  
    </div>  
</form>  
</div>  
</body>  
</html>
```

EXAM QUESTIONS AND ANSWERS

Chapter 2 — HTML and XHTML | Web Technology | Gandaki University

PART A — Short Answer Questions (2 Marks Each)

Q: What is HTML? Write its full form and explain briefly. [2 Marks]

Ans: HTML stands for HyperText Markup Language.

It is the standard language used to create and structure web pages. HTML uses markup tags to describe the structure and presentation of content on the World Wide Web. It is NOT a programming language but a markup language. HTML files are saved with .html or .htm extensions. Tim Berners-Lee invented HTML in 1991. The current version is HTML5.

Q: What is XHTML? How is it different from HTML? [2 Marks]

Ans: XHTML stands for eXtensible HyperText Markup Language.

It is a stricter, XML-based reformulation of HTML. Key differences: In XHTML all tags must be lowercase, all tags must be closed, all attribute values must be in quotes, empty elements must self-close (e.g., `<br />`), and DOCTYPE is mandatory. In HTML, these rules are optional and browsers are lenient with errors.

Q: What are void/empty elements in HTML? Give any four examples. [2 Marks]

Ans: Void elements (also called empty elements or self-closing tags) are HTML elements that do not have any content between an opening and closing tag. They do not have a closing tag. Examples: `<br />` (line break), `<hr />` (horizontal rule), `<img />` (image), `<input />` (form input), `<meta />` (metadata), `<link />` (external link). In XHTML, void elements must always be self-closed with a forward slash: `<br />`.

Q: What is the difference between `<b>` and `<strong>` tags? [2 Marks]

Ans: `<b>` is a presentational element — it makes text visually bold but carries NO semantic meaning. Browsers and screen readers treat it as purely visual.

`<strong>` is a phrase/semantic element — it makes text bold AND communicates that the text has strong importance. Screen readers read `<strong>` content with more emphasis, which is important for accessibility. In HTML5, `<strong>` is preferred over `<b>`.

Q: What is the purpose of the alt attribute in the `<img>` tag? [2 Marks]

Ans: The alt (alternative text) attribute provides a text description of an image.

Its purposes are: (1) Accessibility — screen readers read the alt text to visually impaired users. (2) Fallback — if the image fails to load, the alt text is displayed instead. (3) SEO — search engines use alt text to understand image content. (4) Requirement — alt is mandatory in XHTML. Good practice: alt='Gandaki University main building' rather than alt='image'.

Q: Differentiate between colspan and rowspan attributes. [2 Marks]

Ans: colspan: Makes a cell span horizontally across multiple columns. Example: <td colspan='3'> makes a cell occupy 3 columns side by side.

rowspan: Makes a cell span vertically across multiple rows. Example: <td rowspan='3'> makes a cell occupy 3 rows stacked vertically. colspan merges cells left-right; rowspan merges cells top-bottom.

Q: What is the difference between GET and POST methods in forms? [2 Marks]

Ans: GET: Form data is appended to the URL as a query string (page.php?name=Ram). It is visible in the address bar, limited in size (~2KB), less secure, and suitable for search forms.

POST: Data is sent in the HTTP request body — not visible in the URL. No size limit, more secure, suitable for login/registration forms, file uploads, and sensitive data.

Q: What is the enctype attribute? When is it used? [2 Marks]

Ans: The enctype (encoding type) attribute in the <form> tag specifies how form data is encoded before sending to the server.

Values: (1) application/x-www-form-urlencoded — default, used for text data. (2) multipart/form-data — MUST be used when the form includes file uploads. (3) text/plain — sends plain text. The most important use is: whenever a form has <input type='file'>, set enctype='multipart/form-data' otherwise only the filename, not the file, is sent.

Q: What are phrase elements? Name any four with their purpose. [2 Marks]

Ans: Phrase elements are semantic HTML elements that give meaning to the text they wrap, beyond just visual formatting. They convey information to browsers, search engines, and assistive technologies.

Examples: (strong importance), (emphasis/stress), <abbr> (abbreviation — e.g., <abbr title='HyperText Markup Language'>HTML</abbr>), <mark> (highlighted/relevant text), (deleted content), <ins> (inserted content), <code> (computer code), <cite> (citation/reference).

Q: What is the target attribute in the <a> tag? List its values. [2 Marks]

Ans: The target attribute specifies where to open the linked document when a hyperlink is clicked.

Values: `_self` — opens in the same tab/window (DEFAULT). `_blank` — opens in a new browser tab or window. `_parent` — opens in the parent frame. `_top` — opens in the full browser window (breaks out of frames). `framename` — opens in a named `<iframe>` or frame.

PART B — Long Answer Questions (5–10 Marks)

Q: Explain the complete structure of an HTML document. Write a sample HTML page with all major structural elements. [5 Marks] [5 Marks]

Ans: An HTML document has the following structure:

1. `<!DOCTYPE html>` — Document type declaration. First line of every HTML5 document. Tells browser the document is HTML5.
2. `<html lang='en'>` — Root element wrapping the entire document. Contains head and body.
3. `<head>` — Metadata section. Contains: `<title>` (browser tab title), `<meta charset='UTF-8'>` (character encoding), `<meta name='viewport'>` (mobile scaling), `<link>` (CSS stylesheet), `<script>` (JavaScript).
4. `<body>` — Contains all visible content: headings, paragraphs, images, links, tables, forms.
5. Content elements inside body: `<h1>`–`<h6>` headings, `<p>` paragraphs, `<a>` links, `<img>` images, `<table>` tables, `<form>` forms, `<div>` and `<span>` for grouping.

Example page: `<!DOCTYPE html><html><head><meta charset='UTF-8'><title>Gandaki University</title></head><body><h1>Welcome to Web Technology</h1><p>This is a complete HTML document.</p></body></html>`

Q: Write a complete HTML page demonstrating all types of lists: ordered, unordered, definition, and nested lists. [5 Marks] [5 Marks]

Ans: Three main list types in HTML:

1. Unordered List (`<ul>`) — for items with no specific order. Items shown with bullets. Types: disc, circle, square.
2. Ordered List (`<ol>`) — for sequential items. Items shown with numbers or letters. type attribute: 1, A, a, I, i. start attribute sets starting number.
3. Definition List (`<dl>`) — for term-definition pairs. `<dt>` for term, `<dd>` for definition.
4. Nested Lists — lists can be placed inside `<li>` elements to create multilevel hierarchical lists. All list items use the `<li>` tag except definition lists which use `<dt>` and `<dd>`. CSS is preferred over the type attribute for modern styling.

Q: Explain tables in HTML with all major attributes. Create an HTML table for a timetable using `colspan` and `rowspan`. [10 Marks] [10 Marks]

Ans: HTML tables use: `<table>` (container), `<tr>` (row), `<th>` (header cell — bold+centered), `<td>` (data cell), `<caption>` (table title), `<thead>`, `<tbody>`, `<tfoot>` (semantic row groups).

Key attributes: `border` (border size), `cellpadding` (space inside cells), `cellspacing` (space between cells), `width` (table width), `align` (alignment).

colspan: Merges cell across multiple columns horizontally. Example: `<th colspan='5'>Monday Schedule</th>` creates one header spanning 5 columns.

rowspan: Merges cell across multiple rows vertically. Example: `<td rowspan='3'>8:00–9:00</td>` creates one time slot covering 3 course rows.

Combined use: A cell can use both colspan and rowspan simultaneously.

Best Practice: Use `<thead>`, `<tbody>`, `<tfoot>` for semantic structure. Do NOT use tables for page layout — use CSS instead.

Q: Explain all HTML form controls with syntax and examples. Design a complete Login and Registration form. [10 Marks] [10 Marks]

Ans: Form controls and their purposes:

`<input type='text'>` — Single-line text. `<input type='password'>` — Masked text.

`<input type='email'>` — Email with format validation. `<input type='number'>` — Numeric input.

`<input type='radio'>` — Single selection from group (same name). `<input type='checkbox'>` — Multiple selection.

`<select>` with `<option>` — Dropdown list. multiple attribute allows multi-selection.

`<textarea>` — Multi-line text. rows and cols set visible size.

`<input type='file'>` — File upload. Requires `enctype='multipart/form-data'` on form.

`<input type='hidden'>` — Invisible field to store data.

`<input type='submit'>` — Submits form. `<input type='reset'>` — Clears form.

`<button type='button'>` — Clickable button for custom JS actions.

HTML5 additions: `type='date'`, `'time'`, `'range'`, `'color'`, `'tel'`, `'url'`, `'search'`.

Important attributes: required (mandatory), placeholder (hint), maxlength, min, max, pattern (regex), autocomplete, disabled, readonly.

Q: Compare HTML and XHTML in detail. What are the main rules of XHTML? [5 Marks] [5 Marks]

Ans: HTML vs XHTML — Key Differences:

1. Case sensitivity: HTML tags can be upper/lowercase. XHTML tags MUST be lowercase.
 2. Closing tags: HTML allows unclosed tags (`<br>`, `<p>text`). XHTML MUST close all tags (`<br />`, `<p>text</p>`).
 3. Attribute values: HTML allows unquoted values (`width=200`). XHTML MUST quote all values (`width='200'`).
 4. Attribute minimization: HTML allows `checked`. XHTML requires `checked='checked'`.
 5. DOCTYPE: Optional in HTML. MANDATORY in XHTML.
 6. XML validity: HTML is not XML. XHTML IS valid XML — parseable by XML processors.
 7. XHTML is stricter — errors cause parsing failures unlike HTML which is lenient.
- XHTML was introduced to make web documents consistent across devices and parseable by XML tools.

Q: Explain the <audio> and <video> tags in HTML5. How are they better than Flash? Write complete code examples. [5 Marks] [5 Marks]

Ans: HTML5 <audio> tag: Embeds audio without plugins. Supports MP3, OGG, WAV formats. Attributes: controls (shows player), autoplay, loop, muted, preload. Multiple <source> tags provide fallback formats.

HTML5 <video> tag: Embeds video without plugins. Supports MP4, WebM, OGG. Same attributes as audio plus: width, height, poster (thumbnail image).

Advantages over Flash: (1) No plugin needed — native browser support. (2) Works on iOS/Android — Flash never worked on iPhone. (3) More secure — Flash had major vulnerabilities. (4) Adobe ended Flash in 2020 — HTML5 is the current standard. (5) Better accessibility and SEO. (6) Simpler code — just <audio> or <video> tags.

The <source> element provides multiple format options — browser picks first compatible format.

PART C — MCQ / Objective Questions

Q#	Question	Answer
1	What does HTML stand for?	HyperText Markup Language
2	Who invented HTML?	Tim Berners-Lee (1991)
3	Which tag creates a hyperlink?	<a>
4	Which attribute specifies the link destination?	href
5	Which target value opens a link in a new tab?	_blank
6	Which attribute provides alternative text for images?	alt
7	Which tag creates an unordered (bulleted) list?	
8	Which tag creates an ordered (numbered) list?	
9	Which tag defines a list item?	
10	In a definition list, what does <dt> stand for?	Definition Term
11	Which attribute merges cells horizontally (across columns)?	colspan
12	Which attribute merges cells vertically (across rows)?	rowspan
13	Which tag is a table header cell (bold, centered)?	<th>
14	Which form method shows data in the URL?	GET
15	Which form method is more secure for sensitive data?	POST
16	Which enctype is needed for file uploads?	multipart/form-data
17	Which input type creates a masked (hidden) text field?	type='password'
18	Which input allows only ONE selection from a group?	type='radio'
19	Which input allows MULTIPLE selections?	type='checkbox'

Q#	Question	Answer
20	Which tag creates a multi-line text input?	<textarea>
21	Which tag creates a dropdown list?	<select>
22	Which attribute makes a field mandatory?	required
23	XHTML stands for?	eXtensible HyperText Markup Language
24	XHTML is based on which language?	XML (eXtensible Markup Language)
25	In XHTML, tags must be written in?	Lowercase
26	Which element is the semantic alternative to ?	
27	Which element is the semantic alternative to <i>?	
28	Which HTML5 tag embeds video?	<video>
29	Which HTML5 tag embeds audio?	<audio>
30	Which attribute shows the audio/video control buttons?	controls
31	Which tag provides multiple format options for media?	<source>
32	Which attribute in <input type='file'> allows multiple files?	multiple
33	Which attribute specifies the URL for form submission?	action
34	What is the default method of a form if not specified?	GET
35	Which tag defines the title of a table?	<caption>

Chapter 2 — Summary & Quick Revision

Key Points to Remember

- HTML = HyperText Markup Language. Invented by Tim Berners-Lee, 1991. Current version: HTML5.
- XHTML = HTML + XML rules. Strict: lowercase tags, all closed, quoted attributes, DOCTYPE required.
- HTML document structure: DOCTYPE → <html> → <head> (metadata) → <body> (content).
- Void elements (no closing tag):
, <hr>, , <input>, <meta>, <link>. XHTML needs
.
- Presentational elements (,<i>) are visual only. Phrase elements (,) carry semantic meaning.
- Three list types: (bullets), (numbers), <dl> (term+definition). All use except <dl>.
- Links: — target='_blank' opens new tab; href='mailto:' for email; href='#id' for anchor.
- Images: — alt is required for accessibility and XHTML.
- HTML5 audio: <audio controls><source src='file.mp3'></audio>. No Flash needed.
- HTML5 video: <video controls><source src='file.mp4'></video>. poster attribute shows thumbnail.
- Tables: <table><tr><th><td>. thead/tbody/tfoot for semantic grouping.
- colspan merges cells HORIZONTALLY. rowspan merges cells VERTICALLY.
- Forms: action (where to send), method GET/POST, enctype (multipart for file upload).
- GET: data in URL, bookmarkable, less secure. POST: data in body, secure, for sensitive data.
- Checkboxes: multiple selections. Radio buttons: single selection per name group.
- enctype='multipart/form-data' is REQUIRED for any form with file upload.

Common Exam Mistakes — Avoid These!

WRONG (Avoid)	CORRECT
 or (uppercase)	 (lowercase in XHTML)
 (no alt)	
 for important text	 for important text

WRONG (Avoid)	CORRECT
<code><i></code> for emphasis	<code></code> for emphasis
Using <code>colspan</code> for row merge	Use <code>rowspan</code> for row merge
Forgetting <code>enctype</code> for file upload	<code>enctype='multipart/form-data'</code>
GET for login form (password in URL!)	POST for login/registration forms
Radio buttons with different names	Radio buttons must share the same name
<code><checked></code> in XHTML	<code>checked='checked'</code> in XHTML
Flash for audio/video in HTML5	<code><audio></code> and <code><video></code> tags in HTML5

Best of Luck for Your Examination!

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